```
>>> print(s)
set()          #represents an empty set object
```

- Python supports various mathematical set operations like `union, intersection, subset, superset` etc.

`union(set)`: This method is called on a set object, with the other set as a parameter. The result is the union of the two sets.

```
>>> set1 = {1, 2, 3, 4}
>>> set2 = {4, 5, 6, 7}

>>> set1.union(set2)
{1, 2, 3, 4, 5, 6, 7}
```

`intersection(set)`: This method returns the intersection of the called set and the passed set.

```
>>> set1.intersection(set2)
{4}
```

`issubset(set)`: This method returns a True value if the called set is a subset of the passed set. If we wish to do a check of `x is a proper subset of y`, we should use "`<`" operator.

```
>>> set1 = {1, 2, 3, 4}
>>> set2 = {1, 2}

>>> set2.issubset(set1)
True
>>> set1 > set2 #Tests for a proper subset
True
>>> set2 > set2 #A set is never a proper subset of itself
False
```

`issuperset(set)`: This method checks if the called set is a superset of the passed set. For a proper subset evaluation, we should use the "`>`" operator.

```
>>> set1.issuperset(set2)
True
```